

YICHANG, China

WMO: 574610

Lat: 30.7337N

Lon: 111.3596E

Elev: 845

StdP: 14.25

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 29.9	(c) 31.9	(d) 14.5	(e) 11.8	(f) 40.9	(g) 19.0	(h) 14.7	(i) 40.5	(j) 9.9	(k) 47.6	(l) 9.1	(m) 46.1	(n) 3.0	(o) 110	(p) 0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.3	(c) 96.3	(d) 79.8	(e) 94.1	(f) 78.9	(g) 91.8	(h) 77.9	(i) 82.5	(j) 91.6	(k) 81.4	(l) 90.1	(m) 80.3	(n) 88.5	(o) 6.0	(p) 140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.2	161.9	87.8	79.1	156.0	86.5	78.1	150.8	85.1	47.4	92.0	46.1	90.3	44.8	88.8	92.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.0	8.9	7.6	DB	27.4	100.3	2.2	1.7	25.9	101.6	24.6	102.6	23.4	103.6	21.8	104.8	
			WB	25.0	84.7	2.5	2.2	23.2	86.3	21.7	87.5	20.3	88.8	18.5	90.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	41.5	45.7	54.6	64.5	72.3	78.4	83.0	82.1	75.0	65.6	55.4	45.6
	DBStd	15.35	5.48	6.68	7.24	6.72	5.39	4.91	4.46	4.81	5.46	5.73	6.63	5.26
	HDD50	631	267	149	35	1	0	0	0	0	0	0	25	154
	HDD65	2647	728	540	330	86	8	0	0	0	2	61	292	600
	CDD50	5649	5	29	178	437	691	851	1023	995	749	485	187	19
	CDD65	2190	0	0	8	72	234	401	558	531	301	81	4	0
	CDH74	20823	0	1	68	538	1709	3674	6343	5836	2249	384	21	0
	CDH80	8463	0	0	12	150	573	1417	2881	2599	759	71	1	0
(14) Wind	WSAvg	3.3	3.0	3.2	3.6	3.6	3.5	3.5	3.6	3.5	3.2	2.9	3.0	3.0
(15) Precipitation	PrecAvg	45.1	0.9	1.6	2.3	3.9	5.6	6.1	8.2	6.8	4.0	3.0	2.0	0.7
	PrecMax	63.0	2.4	4.7	4.6	10.6	11.5	10.0	17.7	14.3	9.3	7.6	4.8	1.9
	PrecMin	34.2	0.1	0.2	0.7	0.6	2.1	2.6	2.8	2.0	0.2	0.3	0.1	0.0
	PrecStd	6.9	0.6	1.2	1.1	2.0	2.0	2.1	3.3	3.3	2.5	1.8	1.2	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	61.9	69.4	80.8	89.6	92.9	96.1	99.6	98.9	94.9	86.6	77.1	65.5
		MCWB	48.7	54.0	63.6	67.6	70.9	76.9	80.2	80.3	78.0	68.6	60.1	51.1
	2%	DB	56.2	63.2	74.7	84.3	88.9	93.0	96.8	95.8	90.4	81.4	71.9	60.5
		MCWB	46.2	51.4	60.0	66.7	70.3	76.1	80.9	79.7	75.3	65.7	59.4	49.3
	5%	DB	52.6	59.1	70.2	80.2	86.2	90.4	94.6	93.5	87.3	78.2	68.3	56.8
		MCWB	44.4	49.8	57.6	65.5	69.6	75.7	80.5	79.0	73.3	64.5	58.0	48.0
	10%	DB	49.7	55.6	66.1	76.2	83.0	87.7	91.9	91.0	84.4	75.1	64.9	53.6
		MCWB	43.0	48.0	56.0	63.9	69.1	74.9	79.8	78.2	71.6	64.1	56.8	46.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	57.6	67.1	73.2	77.0	81.7	84.6	83.7	80.7	72.8	64.0	53.8
		MCDB	57.8	64.2	76.2	81.3	85.1	90.3	93.8	93.4	91.1	78.2	68.7	58.4
	2%	WB	47.8	53.4	62.3	70.1	74.6	80.0	82.9	82.4	78.7	69.3	61.9	51.3
		MCDB	53.6	59.6	69.8	78.2	82.0	88.5	92.1	91.3	86.6	76.2	67.6	57.5
	5%	WB	45.8	51.4	59.4	67.7	72.8	78.7	81.8	81.1	76.9	67.3	60.1	49.2
		MCDB	51.5	56.5	67.0	75.8	80.8	86.1	90.8	89.5	83.8	74.1	66.2	55.5
	10%	WB	43.9	49.4	57.1	65.3	71.1	77.3	80.7	80.0	75.0	65.7	58.0	47.4
		MCDB	48.6	54.2	64.3	73.7	78.8	83.4	89.4	87.7	80.7	71.8	63.8	52.8
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.4	13.7	14.8	14.6	13.2	13.3	13.3	12.5	12.3	11.6	10.8
		MCDBR	17.0	19.5	21.5	23.1	21.5	18.2	17.6	17.5	17.4	18.6	18.0	18.3
		MCWBR	8.7	9.5	9.2	8.5	7.3	5.8	5.5	5.2	5.2	6.2	7.3	8.9
	5% WB	MCDBR	14.8	15.1	18.8	19.2	15.8	14.5	15.9	15.5	14.5	13.5	14.6	15.3
		MCWBR	8.2	8.5	9.1	9.0	7.1	5.7	6.2	5.7	5.7	6.0	6.7	8.0
(40) Clear-Sky Solar Irradiance	taub		0.618	0.676	0.689	0.680	0.645	0.669	0.630	0.667	0.683	0.672	0.650	0.602
	taud		1.577	1.508	1.522	1.583	1.699	1.688	1.802	1.700	1.635	1.590	1.589	1.628
	Ebn at Noon		183	188	201	211	220	214	222	212	200	187	174	178
	Edh at Noon		72	84	88	86	77	77	69	75	78	76	70	65
(44) All-Sky Solar Radiation	RadAvg		626	723	1052	1247	1312	1380	1515	1408	1108	905	736	622
	RadStd		113	163	73	142	161	168	174	147	132	134	111	112

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-0.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	+0.76	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page